

Shupeng Chai



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Research interests

Rock Mechanics; Rock Physics; Fault Mechanics; Geotechnical Engineering

Professional Experience

2026–Present	Research Associate	Hong Kong Polytechnic University
2023–2025	Teaching Assistant	Hong Kong Polytechnic University
	• Course tutorial, consultation; Supervision of final year project for undergraduate students	
2021–2023	Lecturer (Assistant)	Zhengzhou University of Science and Technology
	• Teaching (Soil Mechanics, Road Engineering) and research (Slope stability analyses)	
2018	Road Design Intern	Three Gorges Geotechnical Consultants Co., Ltd (Wuhan)

Education

2023–2026	PhD in Geotechnical Engineering	HK Hong Kong Polytechnic University
	• Thesis: Elucidating the Role of Stress Heterogeneity in Fault Shear Behavior through Experimental and Numerical Modeling	
	• Supervisor: Professor Qi Zhao	
2025–2026	Visiting PhD student	SG Nanyang Technological University
	• Project: Simulating the effects of asperity contacts on laboratory earthquakes with PyQuake3D	
	• Supervisor: Professor Luca Dal Zilio	
2018–2020	MASc in Mineral Engineering	CA University of Montreal (Polytechnique Montréal)
	• Thesis: Analytical and numerical studies on the stresses in backfilled stopes and the stability of side-exposed backfill in inclined stopes (link).	
2014–2018	BEng in Civil Engineering	CN Wuhan University
	• Thesis: Numerical analyses on the horizontal directional drilling under a railway (Outstanding Thesis).	

Publications

(To be) submitted

- [1] **Chai S**, Su B, Zou Y, Dal Zilio L, Hatzor Y H, Zhao Q (2026). Fault roughness and contact evolution control dilation and contraction during shear sliding. *Submitted to Journal of Geophysical Research: Solid Earth*.
- [2] **Chai S**, Dal Zilio L, Hatzor Y H, Xia K, Zhao Q (2026). Multi-stage coseismic slip controlled by roughness-induced stress heterogeneity. *To be submitted to Geophysical Research Letters*.
- [3] **Chai S**, Zhao Q (2026). Interplay between fault geometric roughness and mechanical properties in governing sliding instability. *To be submitted to Earth and Planetary Science Letters*.
- [4] **Chai S**, Zou Y, Zhang C, Chen G, Zhao Q (2026). Directional crater landslides induced by moonquakes. *To be submitted Journal of Geophysical Research: Planets*.

Published (selected, see full publication list via Google Scholar)

- [5] **Chai S**, Zou Y, Wu H, Akbariforouz M, Su B, Grasselli G, Elsworth D, Hatzor Y H, Zhao Q* (2026). The role of stress heterogeneity in seismic slip revealed through experiments on laboratory faults. *Rock Mechanics and Rock Engineering*. ([link](#))
- [6] **Chai S**, Zou Y, Wu H, Akbariforouz M, Su B, Grasselli G, Elsworth D, Hatzor YH, Zhao Q* (2026). Influence of stress heterogeneity on shear behavior of rock discontinuities in laboratory experiments: New insights from numerical simulations. *International Journal of Rock Mechanics and Mining Sciences* 197:106358. ([link](#))

- [7] **Chai S**, Zheng J*, and Li L (2023). Kink effect on the stress distribution in 2D backfilled stopes. *Geotechnical and Geological Engineering*. ([link](#))
- [8] **Chai S*** (2023). Two-wedge slope stability analysis considering a nonvertical wedge interface. *Bulletin of Engineering Geology and the Environment*. 82:89. ([link](#))
- [9] Li X, Fan L and **Chai S*** (2022). An experimental study on strengthening recycled aggregates considering dry mixing before slurry coating. *Coatings*. 12(11):1702. ([link](#)).
- [10] **Chai S***, Fan L and Liang H (2022). Required jacking force for deviation rectification of inclined structures supported with rigid piles. *Frontiers in Earth Science*. 10:998798. ([link](#))
- [11] **Chai S*** (2022). Maximum height estimation for mine waste dumps. *Journal of the Southern African Institute of Mining and Metallurgy*. 122(10):579-586. ([link](#))

Selected Conference Presentations

- [1] **Chai S**, Dal Zilio L, Zhao Q (2026). The Dilatancy-Compaction Switch during Earthquakes: Dynamically Governed by the Evolution of Asperity Contacts. **Oral** presentation at *RocDyn-5*.
- [2] **Chai S**, Su B, Zou Y, Zhao Q (2025). Dilation or compaction? Laboratory insights into the role of fault roughness. **Oral** presentation at *EGU General Assembly 2025*. ([link](#)).
- [3] **Chai S**, Zou Y, Chen G, Zhao Q (2024). Possible moonquakes and tectonic activities inferred from crater landslides. **Poster** presentation at *AGU24*. ([link](#))
- [4] **Chai S**, and Zhao Q. (2024). New insights into stress conditions on rock discontinuities in laboratory shear tests. **Oral** presentation at *International Geomechanics Conference 2024*. Kuala Lumpur, Malaysia. (Won **Best Student Award**, [link](#))
- [5] **Chai S**, and Zhao Q. (2024). New insights for stress conditions of laboratory shear tests. **Poster** presentation at *ARMA 58th. US Rock Mechanics/Geomechanics Symposium*. ([link](#))

Awards & Honors

2025	PolyU Research Student Attachment Program (Scholarship) for a 6-month exchange at NTU
2024	Best Student Award at 2024 International Geomechanics Conference
2024	Second Prize in the Student Contest at 2024 International Geomechanics Conference
2024	2024 International Geomechanics Conference Student Sponsorship
2023	Best Poster Award in 2023 ARMA East Asia Geomechanics Workshop
2020	RBC Royal Bank Excellence Scholarship (University of Montreal)
2020	Marianne-Mareschal Excellence Scholarship (University of Montreal)
2019	Quebec Government Exemption Scholarship Program (University of Montreal)
2017	National Encouragement Scholarship (Wuhan University)
2017	Third prize in 11 th National Structure Design Competition
2017	First prize in 10 th National Engineering Drawing and BIM Innovation Competition
2016	Scholarship for Excellence of Wuhan University (Twice)
2016	Outstanding Student Cadre of Wuhan University
2014	Excellent Volunteer, Advanced Individual (Wuhan University)

Skills

Experimental skills	Laboratory shear tests, Acoustic emissions, Micro-CT scan, etc.
Computational skills	FLAC(3D), PFC(3D), MATLAB, Python
Languages	English (Proficient), French (Beginner), Mandarin Chinese (Native)